

Where Two Lines Meet

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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| 1. the pair $(x, y) = x = 4, y = 1$ | 6. intersection point $= x = 5, y = 6$ |
| 2. the pair $(x, y) = x = 4, y = 3$ | 7. (a) the actual solution $= x = 2, y = 5, —$ |
| 3. (a) intersection point $= x = 3, y = 4, —$ | 8. (a) plan J cost at 3 gigabytes $= 38$, (b) the shared (gigabytes, cost) pair $= x = 3, y = 38$ |
| 4. (a) plan A cost at 4 hours $= 24$, (b) plan B cost at 4 hours $= 24$, (c) the shared (hours, cost) pair $= h = 4, y = 24$ | 9. (a) solution, or none $= \emptyset, —$ |
| 5. (a) solution, or none $= \emptyset, —$ | 10. (a) the (visits, cost) pair $= v = 10, y = 50, —$ |
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What is verified

5 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 3, 5, 7, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.EE.C.8a – *problems 1, 2, 3, 4, 6, 8, 9*

8.EE.C.8b – *problems 5, 7*

8.EE.C.8c – *problem 10*

Difficulty 1–4 of 5 across 18 tagged checks.

1. $x + y = 5$ and $x - y = 3$. Find the pair that makes both true.

Solution: Adding the two equations removes y : $2x = 8$, so $x = 4$. Substituting back into $x + y = 5$ gives $y = 1$. Both equations check: $4 + 1 = 5$ and $4 - 1 = 3$. $(4, 1)$

2. $2x + y = 11$ and $x - y = 1$. Find the pair that makes both true.

Solution: Adding removes y : $3x = 12$, so $x = 4$. Then $4 - y = 1$ gives $y = 3$. Check the first equation too: $2(4) + 3 = 11$. $(4, 3)$

3. Lines A ($y = x + 1$) and B ($y = 7 - x$). (a) Crossing point. (b) What it means.

Solution (a): On the grid the two lines meet one gridline right of 2 and level with 4. Algebraically, $x + 1 = 7 - x$ gives $2x = 6$, so $x = 3$ and $y = 4$. $(3, 4)$

Solution (b) — instructor-judged. The point lies on line A and on line B at the same time, so its coordinates satisfy $y = x + 1$ and $y = 7 - x$ together — and that is exactly

what "solution of the system" means. Substituting $(3, 4)$ into both equations is a complete explanation. "It is where they cross" with no link back to the equations is not.

4. Bike plans $y = 4h + 8$ and $y = 2h + 16$, with tables. (a) Plan A at 4 hours. (b) Plan B at 4 hours. (c) The shared pair.

Solution (a): Read the last column of plan A's table.

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Solution (b): Read the last column of plan B's table.

24

Solution (c): The tables agree at 4 hours and \$24, and the algebra agrees: $4h + 8 = 2h + 16$ gives $2h = 8$, so $h = 4$ and $y = 24$.

(4, 24)

5. Gym P: \$25 plus \$8 a class. Gym Q: \$40 plus \$8 a class. (a) When do they cost the same? (b) What that means.

Solution (a): Setting $8c + 25 = 8c + 40$ and subtracting $8c$ leaves $25 = 40$, which is false, so no number of classes makes the costs equal.

no solution

Solution (b) — instructor-judged. Both gyms charge the same \$8 per class, so the \$15 difference in joining fees never closes: gym P is always exactly \$15 cheaper. Graphically the two cost lines have equal slopes and different intercepts, so they are parallel and never meet. "No solution" alone, with no reference to the gyms, is not a complete answer.

6. Lines A ($y = x + 1$) and C ($y = 2x - 4$). Crossing point.

Solution: Reading the grid, the lines meet at the corner five right and six up. Algebraically $x + 1 = 2x - 4$ gives $x = 5$, and then $y = 6$.

(5, 6)

7. A student claims $(3, 2)$ solves $3x + y = 11$ and $x + 2y = 12$. (a) The actual solution. (b) Why the claim fails.

Solution (a): From the first equation $y = 11 - 3x$. Substituting into the second: $x + 2(11 - 3x) = 12$, so $x + 22 - 6x = 12$, giving $-5x = -10$ and $x = 2$; then $y = 5$.

(2, 5)

Solution (b) — instructor-judged. Substituting $(3, 2)$ into the first equation gives $3(3) + 2 = 11$, which is true — and that is exactly why the claim is tempting. The second equation gives $3 + 2(2) = 7$, not 12, so the pair fails there. A solution of a system must satisfy every equation, so one hit out of two is not enough. Full credit needs both substitutions shown, or at least the failing one named.

8. Plan J: $y = 6x + 20$ with a table; plan K: $y = 2x + 32$. (a) Plan J at 3 gigabytes. (b) The shared pair.

Solution (a): Read the last column of the table.

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Solution (b): Set the two costs equal: $6x + 20 = 2x + 32$ gives $4x = 12$, so $x = 3$, and then $y = 6(3) + 20 = 38$ — the same 38 the table shows.

(3, 38)

9. Lines A ($y = x + 1$) and D ($y = x + 4$). (a) Solution or none. (b) How the graph shows it.

Solution (a): Setting $x + 1 = x + 4$ and subtracting x leaves $1 = 4$, which is false.

no solution

Solution (b) — instructor-judged. On the Graph Bank, lines A and D climb at the same rate, so the vertical gap between them stays at 3 units all the way across the grid and they never meet. A system whose graphs never cross has no pair that lies on both, so it has no solution. Saying only "they are parallel" does not say what that means for the system.

10. Pool: $y = 5v$ against $y = 2v + 30$. (a) Where they cost the same. (b) What it means for a swimmer.

Solution (a): $5v = 2v + 30$ gives $3v = 30$, so $v = 10$, and the cost is $5(10) = 50$ dollars. $(10, 50)$

Solution (b) — instructor-judged. At exactly 10 visits the two options cost the same, \$50. Below 10 visits the pay-per-visit line is the lower of the two, so single visits are cheaper; above 10 visits the membership line is lower, so the membership is cheaper. A correct answer names the crossing and which option wins on each side of it; the coordinates repeated in words are not enough.